

**Q1.** The correct order of dipole moments for molecules  $\text{NH}_3$ ,  $\text{H}_2\text{S}$ ,  $\text{CH}_4$  and  $\text{HF}$  is: **(2023)**

- (a)  $\text{CH}_4 > \text{H}_2\text{S} > \text{NH}_3 > \text{HF}$
- (b)  $\text{H}_2\text{S} > \text{NH}_3 > \text{HF} > \text{CH}_4$
- (c)  $\text{NH}_3 > \text{HF} > \text{CH}_4 > \text{H}_2\text{S}$
- (d)  $\text{HF} > \text{NH}_3 > \text{H}_2\text{S} > \text{CH}_4$

**Q2.** Which one of the following represents all isoelectronic species? **(2023)**

- (a)  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{O}^-$ ,  $\text{NO}^+$
- (b)  $\text{N}_2\text{O}$ ,  $\text{N}_2\text{O}_4$ ,  $\text{NO}^+$ ,  $\text{NO}$
- (c)  $\text{Na}^+$ ,  $\text{Mg}^{2+}$ ,  $\text{O}^-$ ,  $\text{F}^-$
- (d)  $\text{Ca}^{2+}$ ,  $\text{Ar}$ ,  $\text{K}^+$ ,  $\text{Cl}^-$

**Q3.** Which one of the following statements is incorrect related to Molecular Orbital Theory? **(2023)**

- (a) The  $\pi^*$  antibonding molecular orbital has a node between the nuclei.
- (b) In the formation of bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other.
- (c) Molecular orbitals obtained from  $2p_x$  and  $2p_y$  orbitals are symmetrical around the bond axis.
- (d) A  $\pi$ -bonding molecular orbital has larger electron density above and below the internuclear axis.

**Q4.** Given below are two statements:

Statement I: Hydrated chlorides and bromides of Ca, Sr and Ba on heating undergo hydrolysis.

Statement II: Hydrate chlorides and bromides of Be and Mg on heating undergo dehydration.

In the light of the above statements, choose the correct answer from the options given below: **(2023)**

- (a) Statement I is correct but Statement II is false.
- (b) Statement II is incorrect but Statement II is true.
- (c) Both Statement I and Statement II are true.
- (d) Both Statement I and Statement II are false.

**Q5.** The correct order of energies of molecular orbitals of  $\text{N}_2$  molecule, is **(2023)**

- (a)  $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
- (b)  $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$
- (c)  $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$
- (d)  $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$



**Q6.** Talking stability as the factor, which one of the following represents correct relationship? **(2023)**

- (a)  $\text{InI}_3 > \text{InI}$
- (b)  $\text{AlCl} > \text{AlCl}_3$
- (c)  $\text{TlI} > \text{TlI}_3$
- (d)  $\text{TlCl}_3 > \text{TlCl}$

**Q7.** Intermolecular forces are forces of attraction and repulsion between interacting particles that will include:

- A. dipole-dipole forces
- B. dipole-induced dipole forces
- C. hydrogen bonding
- D. covalent bonding
- E. dispersion forces

Choose the most appropriate answer from the options given below: **(2023)**

- (a) A, B, C, D are correct
- (b) A, B, C, E are correct
- (c) A, C, D, E are correct
- (d) B, C, D, E are correct

**Q8.** Match List I with List II:

| List I<br>(Molecules) | List II<br>(Shape)       |
|-----------------------|--------------------------|
| A. $\text{NH}_3$      | i. Square pyramidal      |
| B. $\text{ClF}_3$     | ii. Trigonal bipyramidal |
| C. $\text{PCl}_5$     | iii. Trigonal pyramidal  |
| D. $\text{BrF}_5$     | iv. T-shape              |

Choose the correct answer from the options given below: **(2022)**

- (a) A-iii, B-iv, C-i, D-ii
- (b) A-ii, B-iii, C-iv, D-i
- (c) A-iii, B-iv, C-ii, D-i
- (d) A-iv, B-iii, C-i, D-ii

**Q9.** The correct order of bond angles in the following compounds/species is: **(2022)**

- (a)  $\text{CO}_2 < \text{NH}_3 < \text{H}_2\text{O} < \text{NH}_4^+$
- (b)  $\text{H}_2\text{O} < \text{NH}_3 < \text{NH}_4^+ < \text{CO}_2$
- (c)  $\text{H}_2\text{O} < \text{NH}_4^+ < \text{NH}_3 < \text{CO}_2$
- (d)  $\text{H}_2\text{O} < \text{NH}_4^+ = \text{NH}_3 < \text{CO}_2$



**Q10.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):  $\text{ICl}$  is more reactive than  $\text{I}_2$ .

Reason (R):  $\text{I-Cl}$  bond is weaker than  $\text{I-I}$  bond.

In the light of the above statements, choose the most appropriate answer from the options given below: **(2022)**

- (a) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (c) (A) is correct but (R) is not correct.
- (d) (A) is not correct but (R) is correct.

**Q11.** Amongst the following which one will have maximum 'lone pair – lone pair' electron repulsions? **(2022)**

- (a)  $\text{ClF}_3$
- (b)  $\text{IF}_5$
- (c)  $\text{SF}_4$
- (d)  $\text{XeF}_2$

**Q12.** Which amongst the following is incorrect statement? **(2022)**

- (a) The bond orders of  $\text{O}_2^+$ ,  $\text{O}_2$ ,  $\text{O}_2^-$  and  $\text{O}_2^{2-}$  are 2.5, 2, 1.5 and 1, respectively
- (b)  $\text{C}_2$  molecule has four electrons in its two degenerate  $\pi$  molecular orbitals
- (c)  $\text{H}_2^+$  ion has one electron
- (d)  $\text{O}_2^+$  ion is diamagnetic

**Q13.**  $\text{BF}_3$  is planar and electron compound. Hybridization and number of electrons around the **central atom**, respectively are : **(2021)**

- (a)  $\text{sp}^3$  and 6
- (b)  $\text{sp}^2$  and 6
- (c)  $\text{sp}^2$  and 8
- (d)  $\text{sp}^3$  and 4

**Q14.** Match List-I with List-II. **(2021)**

| List-I            | List-II                   |
|-------------------|---------------------------|
| A. $\text{PCl}_5$ | (i) Square pyramidal      |
| B. $\text{SF}_6$  | (ii) Trigonal planar      |
| C. $\text{BrF}_5$ | (iii) Octahedral          |
| D. $\text{BF}_3$  | (iv) Trigonal bipyramidal |

Choose the correct answer from the options given below.



- (a) A-ii, B-iii, C-iv, D-i
- (b) A-iii, B-I, C-iv, D-ii
- (c) A-iv, B-iii, C-ii, D-i
- (d) A-iv, B-iii, C-I, D-ii

**Q15.** Which of the following molecules is non-polar in nature? **(2021)**

- (a) CH<sub>2</sub>O
- (b) SbCl<sub>5</sub>
- (c) NO<sub>2</sub>
- (d) POCl<sub>3</sub>

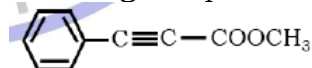
**Q16.** Which of the following set of molecules will have zero dipole moment? **(2020)**

- (a) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene
- (b) Nitrogen trifluoride, beryllium difluoride, water 1, 3-dichlorobenzene
- (c) Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene
- (d) Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene

**Q17.** Identify a molecule which does not exist. **(2020)**

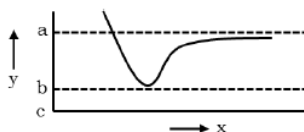
- (a) Li<sub>2</sub>
- (b) C<sub>2</sub>
- (c) O<sub>2</sub>
- (d) He<sub>2</sub>

**Q18.** How many (i) sp<sup>2</sup> hybridised carbon atoms and (ii)  $\pi$  bonds are present in the following compound? **(2020 Covid Re-NEET)**



- (a) 8, 6
- (b) 7, 6
- (c) 8, 5
- (d) 7, 5

**Q19.** The potential energy (y) curve for H<sub>2</sub> formation as a function of internuclear distance (x) of the H atoms is shown below.



The bond energy of H<sub>2</sub> is **(2020 Covid Re-NEET)**

- (a) (c-a)<sup>2</sup>
- (b) (b-a)<sup>2</sup>
- (c) (c-a)
- (d) (b-a)

**Q20.** Identify the wrongly matched pair. **(2020 Covid Re-NEET)**



**Molecule Shape or geometry of molecule**

- (a) SF<sub>6</sub> Octahedral
- (b) BeCl<sub>2</sub> Linear
- (c) NH<sub>3</sub> Trigonal pyramidal
- (d) PCl<sub>5</sub> Trigonal planar

**Q21.** Which of the following diatomic molecular species has only  $\pi$  bonds according to Molecular Orbital Theory? **(2019)**

- (a) O<sub>2</sub>
- (b) N<sub>2</sub>
- (c) C<sub>2</sub>
- (d) Be<sub>2</sub>

**Q22.** Identify the incorrect statement related to PCl<sub>5</sub> from the following: **(2019)**

- (a) Three equatorial P–Cl bonds make an angle of 120° with each other
- (b) Two axial P–Cl bonds make an angle of 180° with each other
- (c) Axial P–Cl bonds are longer than equatorial P–Cl bonds
- (d) PCl<sub>5</sub> molecule is non-reactive

**Q23.** Consider the following species: **(2018)**

CN<sup>+</sup>, CN<sup>–</sup>, NO and CN

Which one of these will have the highest bond order?

- (a) NO
- (b) CN<sup>–</sup>
- (c) CN
- (d) CN<sup>+</sup>

**Q24.** Which one of the following pair of species have the same bond order? **(2017-Delhi)**

- (a) N<sub>2</sub>, O<sub>2</sub><sup>–</sup>
- (b) CO, NO
- (c) O<sub>2</sub>, NO<sup>+</sup>
- (d) CN<sup>–</sup>, CO

**Q25.** The species, having bond angles of 120° is **(2017-Delhi)**

- (a) BCl<sub>3</sub>
- (b) PH<sub>3</sub>
- (c) ClF<sub>3</sub>
- (d) NCl<sub>3</sub>

**Q26.** Which one of the following ions is not tetrahedral in shape? **(2017-Gujarat)**

- (a) [NiCl<sub>4</sub>]<sup>2–</sup>
- (b) NH<sub>4</sub><sup>+</sup>
- (c) BF<sub>4</sub><sup>–</sup>
- (d) [Cu(NH<sub>3</sub>)<sub>4</sub>]<sup>2+</sup>



**Q27.** Which of the following pair of species is not iso-structural? (2017-Gujarat)

- (a)  $BrO_3^-$ ,  $XeO_3$
- (b)  $ICl_4^-$ ,  $XeF_4$
- (c)  $ClO_3^-$ ,  $CO_3^{2-}$
- (d)  $IBr_2^-$ ,  $XeF_2$

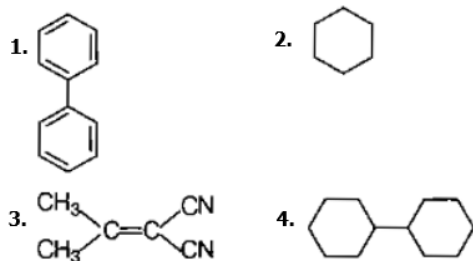
**Q28.** Which of the following hydrides has the largest bond angle? (2017-Gujarat)

- (a)  $H_2Se$
- (b)  $H_2S$
- (c)  $H_2Te$
- (d)  $H_2O$

**Q29.** Which one of the following compounds shows the presence of intramolecular hydrogen bond? (2016-II)

- (a) Cellulose
- (b) Concentrated acid
- (c)  $H_2O_2$
- (d)  $HCN$

**Q30.** In which of the following molecules, all atoms are coplanar?



**Q31.** Among the following which one is a wrong statement? (2016-II)

- (a)  $SeF_4$  and  $CH_4$  have same shape
- (b)  $I_3^+$  has bent geometry
- (c)  $PH_5$  and  $BiCl_5$  do not exist
- (d) p $\pi$  - d $\pi$  bonds are present in  $SO_2$

**Q32.** The hybridisations of atomic orbitals of nitrogen in  $NO_2^+$ ,  $NO_3^-$  and  $NH_4^+$  respectively are : (2016-II)

- (a)  $sp$ ,  $sp^3$  and  $sp^2$
- (b)  $sp^2$ ,  $sp^3$  and  $sp$
- (c)  $sp$ ,  $sp^2$  and  $sp^3$
- (d)  $sp^2$ ,  $sp$  and  $sp^3$

**Q33.** Which of the following pairs of ions is isoelectronic and isostructural? (2016-II)

- (a)  $CO_3^{2-}$ ,  $NO_3^-$



- (b)  $ClO_3^-$ ,  $CO_3^{2-}$
- (c)  $SO_3^{2-}$ ,  $NO_3^-$
- (d)  $ClO_3^-$ ,  $SO_3^{2-}$

**Q34.** Consider the molecules  $CH_4$ ,  $NH_3$  and  $H_2O$ . Which of the given statement is false? **(2016-I)**

- (a) The H-C-H bond angle in  $CH_4$  is larger than the H-N-H bond angle in  $NH_3$
- (b) The H-C-H bond angle in  $CH_4$ , the H-N-H bond angle in  $NH_3$ , and the H-O-H bond angle in  $H_2O$  are all greater than  $90^\circ$
- (c) Then H-O-H bond angle in  $H_2O$  is larger than the H-C-H bond angle in  $CH_4$
- (d) The H-O-H bond angle in  $H_2O$  is smaller than the H-N-H bond angle in  $NH_3$

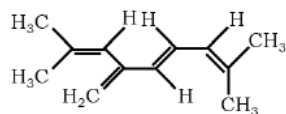
**Q35.** Predict the correct order among the following: **(2016-I)**

- (a) Lone pair – bond pair > bond pair – bond pair > lone pair – lone pair
- (b) Lone pair – lone pair > lone pair – bond pair > bond pair – bond pair
- (c) Lone pair – lone pair > bond pair – bond pair > lone pair – bond pair
- (d) Bond pair – bond pair > lone pair – bond pair > lone pair – lone pair

**Q36.** Decreasing order of stability of  $O_2$ ,  $O_2^-$ ,  $O_2^+$  and  $O_2^{2-}$  is : **(2015 RE)**

- (a)  $O_2^- > O_2^{2-} > O_2^+ > O_2$
- (b)  $O_2^+ > O_2 > O_2^- > O_2^{2-}$
- (c)  $O_2^{2-} > O_2^- > O_2 > O_2^+$
- (d)  $O_2 > O_2^+ > O_2^{2-} > O_2^-$

**Q37.** The total number of  $\pi$ -bond electrons in the following structure is:



- (a) 8
- (b) 12
- (c) 16
- (d) 4

**Q38.** Which of the following options represents the correct bond order? **(2015)**

- (a)  $O_2^- < O_2 < O_2^+$
- (b)  $O_2^- > O_2 < O_2^+$
- (c)  $O_2^- < O_2 > O_2^+$
- (d)  $O_2^- > O_2 > O_2^+$

**Q39.** The correct bond order in the following species is: **(2015)**

- (a)  $O_2^{2+} < O_2^- < O_2^+$
- (b)  $O_2^{2+} < O_2^- < O_2^+$
- (c)  $O_2^- < O_2^+ < O_2^{2+}$
- (d)  $O_2^{2+} < O_2^+ < O_2^-$



**Q40.** Which of the following pairs of ions are isoelectronic and isostructural? (2015)

- (a)  $\text{SO}_3^{2-}$ ,  $\text{NO}_3^-$
- (b)  $\text{ClO}_3^-$ ,  $\text{SO}_3^{2-}$
- (c)  $\text{CO}_3^{2-}$ ,  $\text{SO}_3^{2-}$
- (d)  $\text{ClO}_3^-$ ,  $\text{CO}_3^{2-}$

**Q41.** Maximum bond angle at nitrogen is present in which of the following? (2015)

- (a)  $\text{NO}_2^+$
- (b)  $2\text{NO}_3^-$
- (c)  $\text{NO}_2$
- (d)  $\text{NO}_2^-$

**Q42.** Which of the following molecules has the maximum dipole moment? (2014)

- (a)  $\text{CH}_4$
- (b)  $\text{NH}_3$
- (c)  $\text{NF}_3$
- (d)  $\text{CO}_2$

**Q43.** Which one of the following species has planar triangular shape? (2014)

- (a)  $\text{NO}_3^-$
- (b)  $\text{NO}_2^-$
- (c)  $\text{CO}_2$
- (d)  $\text{N}_3$

**Q44.** Which of the following organic compounds has same hybridization as its combustion product ( $\text{CO}_2$ )? (2014)

- (a) Ethyne
- (b) Ethene
- (c) Ethanol
- (d) Ethane

**Q45.** Which of the following is a polar molecule? (2013)

- (a)  $\text{BF}_3$
- (b)  $\text{SF}_4$
- (c)  $\text{SiF}_4$
- (d)  $\text{XeF}_4$

**Q46.** Which of the following is electron-deficient? (2013)

- (a)  $(\text{CH}_3)_2$
- (b)  $(\text{SiH}_3)_2$
- (c)  $(\text{NH}_3)_2$
- (d)  $\text{PH}_3$





**Q47.** Which of the following is paramagnetic? **(2013)**

- (a) CO
- (b)  $O_2^-$
- (c) CN
- (d)  $NO^+$



## ANSWER KEY

- |              |              |
|--------------|--------------|
| 1. Ans. (d)  | 25. Ans. (a) |
| 2. Ans. (d)  | 26. Ans. (d) |
| 3. Ans. (c)  | 27. Ans. (c) |
| 4. Ans. (d)  | 28. Ans. (d) |
| 5. Ans. (d)  | 29. Ans. (a) |
| 6. Ans. (c)  | 30. Ans. (c) |
| 7. Ans. (b)  | 31. Ans. (a) |
| 8. Ans. (c)  | 32. Ans. (c) |
| 9. Ans. (b)  | 33. Ans.     |
| 10. Ans. (a) | (a/d)        |
| 11. Ans. (d) | 34. Ans. (c) |
| 12. Ans. (d) | 35. Ans. (b) |
| 13. Ans. (b) | 36. Ans. (b) |
| 14. Ans. (d) | 37. Ans. (a) |
| 15. Ans. (b) | 38. Ans. (a) |
| 16. Ans. (c) | 39. Ans. (c) |
| 17. Ans. (d) | 40. Ans. (b) |
| 18. Ans. (b) | 41. Ans. (a) |
| 19. Ans. (d) | 42. Ans. (b) |
| 20. Ans. (d) | 43. Ans. (a) |
| 21. Ans. (c) | 44. Ans. (a) |
| 22. Ans. (d) | 45. Ans. (b) |
| 23. Ans. (b) | 46. Ans. (c) |
| 24. Ans. (d) | 47. Ans. (b) |



## SOLUTION

1. Ans.(d)

$\text{HF} > \text{NH}_3 > \text{H}_2\text{S} > \text{CH}_4$  (non-polar)

2. Ans.(d)

Total numbers electrons are same

$\text{Cr}^{+2}, \text{Ar}, \text{K}^+, \text{Cl}^- \rightarrow 20$  electrons

3. Ans.(c)

In the formation of BMO, the two electron waves of the bonding atoms reinforce each other due to constructive interference. Molecular orbitals obtained from  $2p_x$  and  $2p_y$  orbitals are 'unsymmetrical' around bond axis.

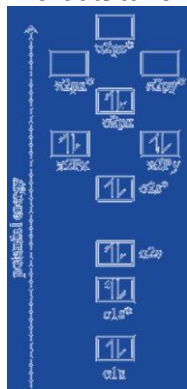
4. Ans.(d)

Hydrated chlorides and Bromides of Ca, Sr and Ba are ionic so undergo dehydration after heating. Hydrated chlorides and Bromides of Be and Mg are covalent so undergo hydrolysis on Heating.

$\text{NH}_3, \text{AlCl}_3, \text{BeCl}_2, \text{CCl}_4, \text{PCl}_5$

5. Ans.(d)

Molecular orbital (energy) diagram sequence of  $\text{N}_2$



6. Ans.(c)

$\text{Tl}^+ \text{ \& } \text{I}^- \rightarrow \text{Tl}^{+3} \text{ \& } 3\text{I}^-$

due to inert pair effect  $\text{Tl}^+$  is more stable than  $\text{Tl}^{+3}$ .

7. Ans.(b)

Intermolecular forces means force of attraction between two or more molecules dipole-dipole (attraction between two or more polar molecules).

Dipole induced dipole (attraction between polar and non-polar molecules)

Hydrogen bonding (it is a special type of dipole-dipole and ion-dipole attraction)

Dispersion forces (mainly acts between non-polar molecules).

Covalent bonding (acts between atom not between molecules).

8. Ans.(c) A-iii, B-iv, C-ii, D-i

9. Ans.(b)

$\text{CO}_2 \Rightarrow \text{sp}^2$  hybridisation, bond angle =  $180^\circ$



$\text{NH}_4^+ \Rightarrow \text{sp}^3$  hybridisation, bond angle =  $109^\circ 28'$

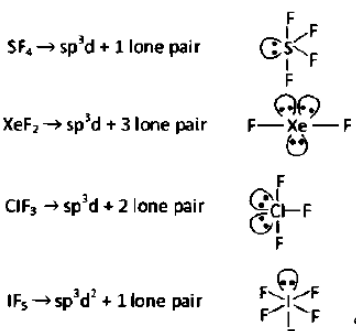
$\text{NH}_3 \Rightarrow \text{sp}^3$  hybridisation with one lone pair on central atom, bond angle  $\approx 107^\circ$

$\text{H}_2\text{O} \Rightarrow \text{sp}^3$  hybridisation with two lone pairs on central atom, bond angle  $\approx 104.5^\circ$

10. Ans.(a)

In general, interhalogen compounds are more reactive than halogens (except fluorine). This is because X-X' bond in interhalogens is weaker than X-X bond in halogens excepts F-F bond. Therefore I-Cl is more reactive than  $\text{I}_2$  because of weaker I-Cl bond then I-I bond.

11. Ans.(d)



$\text{XeF}_2$  having maximum lone pairs, so, it has maximum 'lone pair – lone pair' electron repulsions.

12. Ans.(d)

Due to one unpaired electron in  $\pi^*2p$  molecular orbital,  $\text{O}_2^+$  is a paramagnetic ion.

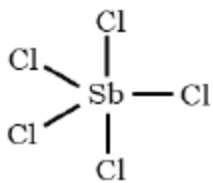
13. Ans.(b)

14. Ans.(d)

|   |                | Hybridisation           | L.P. | Shape                |
|---|----------------|-------------------------|------|----------------------|
| A | $\text{PCl}_5$ | $\text{sp}^3\text{d}$   | 0    | Trigonal bipyramidal |
| B | $\text{SF}_6$  | $\text{sp}^3\text{d}^2$ | 0    | Octahedral           |
| C | $\text{BrF}_5$ | $\text{sp}^3\text{d}^2$ | 1    | Square pyramidal     |
| D | $\text{BF}_3$  | $\text{sp}^2$           | 0    | Trigonal planar      |

15. Ans.(b)

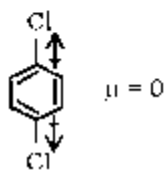
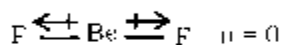
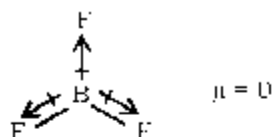




sp<sup>3</sup>d Dipole moment ( $\mu$ ) = 0  
 Trigonal Non-polar  
 Bipyramidal

16. Ans.(c)

The given set of molecules have dipole moment zero. Because dipoles of the bond cancel each other.



17. Ans.(d)

For He<sub>2</sub> molecule

Electronic configuration is  $1s^2, 1s^{*2}$

B.O. =  $12(Nb - Na)$

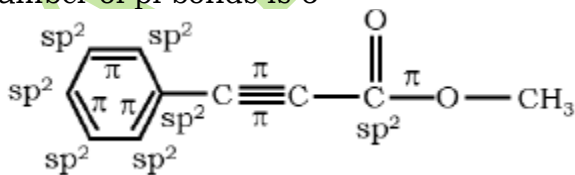
=  $12(2 - 2) = 0$

The bond order comes out to be zero. This indicates that there is no bond formation between 2 He atoms and hence the He<sub>2</sub> molecule does not exist.

18. Ans.(b)

Number of sp<sup>2</sup> hybridised carbon atoms is 7

Number of pi bonds is 6



19. Ans.(d)

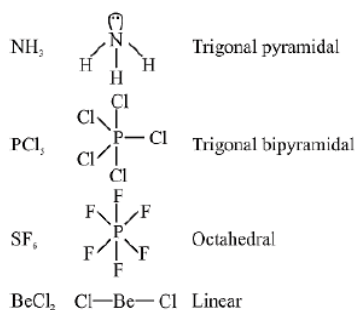
Potential energy of two H atoms at infinite distance = a

Potential energy of two H atoms at distance equal to bond length = b

So, the bond energy of H<sub>2</sub> = (b - a)



20. Ans.(d)



21. Ans.(c)

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2$

Double bond in  $\text{C}_2$  consists of both  $\square$  bonds because of the presence of four electrons in two  $\square$  molecular orbital. In other molecule a double bond is made up of a sigma bond and a pi bond.

22. Ans.(d)

(d) False

Due to longer and thus weaker axial bonds,  $\text{PCl}_5$  is a reactive molecule.

23. Ans.(b)

$\text{NO} : , (\sigma^* 1s)^2, (\sigma 2s)^2, (\sigma^* 2s)^2, (\sigma 2p_z)^2, (\pi 2p_x)^2 = (\pi 2p_y)^2, (\pi^* 2p_x)^1 = (\pi^* 2p_y)^0$

$BO = 10 - 5/2 = 2.5$

$\text{CN}^- : (\sigma 1s)^2, (\sigma^* 1s)^2, (\sigma 2s)^2, (\sigma^* 2s)^2, (\pi 2p_x)^2 = (\pi 2p_y)^2, (\sigma 2p_z)^2$

$BO = 10 - 4/2 = 3$

$\text{CN} : (\sigma 1s)^2, (\sigma^* 1s)^2, (\sigma 2s)^2, (\sigma^* 2s)^2, (\pi 2p_x)^2 = (\pi 2p_y)^2, (\sigma 2p_z)^1$

$BO = 9 - 4/2 = 2.5$

$\text{CN}^+ : (\sigma 1s)^2, (\sigma^* 1s)^2, (\sigma 2s)^2, (\sigma^* 2s)^2, (\pi 2p_x)^2 = (\pi 2p_y)^2$

$BO = 8 - 4/2 = 2$

Hence, option (b) should be the right answer.

24. Ans.(d)

$\text{CN}^- = 6 + 7 + 1 = 14$

$\text{CO} = 6 + 8 = 14$

These two species are isoelectronic and iso structural in nature. Therefore, they have both have same bond order.

25. Ans.(a)

$\text{BCl}_3$  having bond angles of  $120^\circ$ .

26. Ans.(d)

$[\text{Cu}(\text{NH}_3)_4]^{2+}$  is not tetrahedral. It is a square planar complex.

27. Ans.(c)

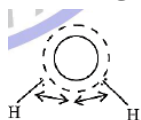


Hence, they both are not isostructural.

28. Ans.(d)

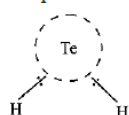
O, Se, Te belong to Group 16.

On moving down the group, size of atom increases.



O is most electronegative and lone pairs lie close to the atom electron cloud. This causes repulsion in lone pairs of oxygen and bond pairs of hydrogen.

∴ Angle maximum due to l.p. – b.p. repulsion.



Te has maximum size: lone pair lie far away from the atom electron cloud. Lone pair – bond pair repulsion is the least.

29. Ans.(a)

Intramolecular hydrogen bonding takes place within the same molecule i.e., between the atoms of the same molecule. Since, cellulose is a complex structure containing oxygen and hydrogen the bonding occurs between them easily.

Whereas, in HCN, H<sub>2</sub>O<sub>2</sub> and concentrated acetic acid intermolecular hydrogen bonding occurs.

Eg.:

30. Ans.(c)

Coplanar are in a plane or where all C atoms are sp<sup>2</sup> hybridised.

31. Ans.(a)

SeF<sub>4</sub> and CH<sub>4</sub> do not have same shape. SeF<sub>4</sub> is AB<sub>4</sub>L type molecule with 4 bond pair and 1 lone pair with shape see-saw. CH<sub>4</sub> is AB<sub>4</sub> type molecule with no lone pair and tetrahedral shape.

I<sub>3</sub><sup>+</sup> have 2 lone pairs with bent/angular shape. BiCl<sub>5</sub> does not exist because of inert pair effect. SO<sub>2</sub> type molecule have both p – p & d – p bonds.

32. Ans.(c)

Hybridisation state = Number of  $\sigma$  bond + number of lone pair

Or

Hybridisation state  $\rightarrow$  from steric number rule

Hybridisation state

$$= \frac{1}{2}(V.E + MA - C + a)$$

$$\text{For, } NO_2 = \frac{1}{2}(5+0-1)$$

$$= 2 \rightarrow sp$$

$$NO_3^- = \frac{1}{2}(5+0+1)$$

$$= 3 \rightarrow sp^2$$



$$NH_4^+ = 12[5+4-1]$$

$$= 4 \rightarrow sp^3$$

33. Ans.(a, d)

34. Ans.(c)

$CH_4$  –  $sp^3$  hybridised, tetrahedral bond angle  $109^\circ 28'$

$NH_3$  –  $sp^3$  hybridised, bond angle –  $107^\circ$

$H_2O$  –  $sp^3$  – bent shaped bond angle –  $104^\circ 5'$

So, bond angle of  $H_2O$  is less than that of  $NH_3$  &  $CH_4$

35. Ans.(b)

Order of repulsing force according to VSEPR theory is lone pair – lone pair > lone pair – bond pair > bond pair – bond pair.

36. Ans.(b)

$O_2$  (atomic number) = 16

Molecular orbital Diagram :

Bond order =  $\frac{1}{2}$  (No. of bonding orbital – no. of anti bonding orbitals)

$$= \frac{1}{2} (10 - 6) = 2$$

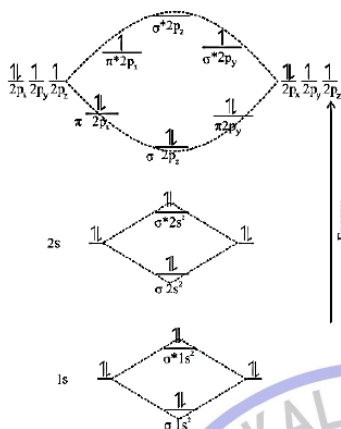
Similarly, For  $O_2^-$  = 1.5

$$O_2^{2-} = 1$$

and Bond order  $\propto$  Bond length  $\propto$  stability

$\therefore$  Order of Bond length

$$= O_2^{2-} > O_2^- > O_2 > O_2^+$$



37. Ans.(a)

Each pi bond have 2e- involved so 4 pi bonds = 8e-.

38. Ans.(a)

Bond order :  $O_2^-$  = 1.5  $O_2$  = 2  $O_2^+$  = 2.5

$$O_2^+ > O_2 > O_2^-$$

39. Ans.(c)

Indention Bond order :

$$O_2^{2+} = 3, O_2^- = 1.5, O_2^+ = 2.5$$



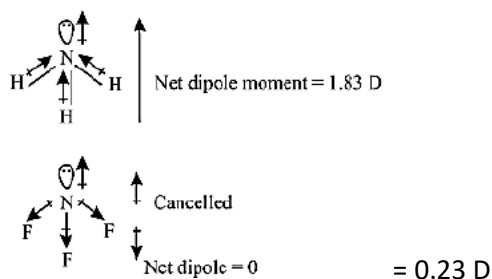


→  $O_2^{2+} + >O_2 + >O_2^-$

40. Ans.(b)

41. Ans.(a)

42. Ans.(b)



43. Ans.(a)

$NO_3^-$  – resonating structure

Planar structure

44. Ans.(a)

Hybridisation of  $CO_2$  is  $sp$   $O = C = O$  and of ethyne  $C_2H_2$  is  $sp$   $H - C \equiv C - H$

45. Ans.(b)

46. Ans.(c)

Among the 4 options  $(CH_3)_2$ ,  $(SiH_3)_2$  and  $PH_3$  all have complete octets with  $8e^-$ . But  $BH_3$  has only  $6e^-$  and act as a Lewis acid.

47. Ans.(b)

In paramagnetic species there is a presence of unpaired electrons.

Except  $O_2^-$  all contains paired electrons

